Individual Project Report

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Graduate Certificate in Pattern Recognition Systems · Practice Module

1. My contribution to the project

V-Trace was a one-person project (Group 21), so I was responsible for every part of it:

- **Problem framing and proposal.** I defined the two-stage *describe → search → confirm → re-identify* workflow, the scope (uploaded videos, ranked results for human review, no identity claims) and the evaluation plan, and presented it in the first presentation.
- **Recognition pipeline.** I built video ingestion with YOLO11 detection and ByteTrack tracking, turned tracks into tracklets with quality-based crop selection, and computed six complementary descriptors per tracklet: ResNet-50 and CLIP embeddings, colour/type probabilities, an HSV histogram, EasyOCR plate text with wildcard matching, and CLIP-scored distinctive cues.
- **Learning components.** I trained a supervised colour/type probe on VeRi-776 labels, a Random Forest same/different-vehicle matcher over a ten-feature evidence vector, and an unsupervised Self-Organizing Map, and compared them against appearance-only and weighted-fusion baselines.
- **Path tracing.** I added a path-growing method that follows a confirmed vehicle from camera to camera and built multi-camera demo footage on which the same cars really appear in three cameras.
- **Evaluation.** I designed an identity-disjoint VeRi-776 protocol, a plate-degradation experiment, ablations, and a pseudo-ground-truth evaluation of tracking and path tracing on the demo cameras.
- **Application and engineering.** I built the Streamlit app (admin sign-in, dark/light theme, six tabs), unit and app tests with GitHub Actions CI, and managed the work through GitHub issues, branches and pull requests.
- **Documentation.** I wrote the final report, both presentations and the demo guide, and recorded the video.

2. What I learned that is most useful to me

Evaluation discipline matters as much as the model. My first Random Forest ranked *worse* than a single appearance cue. The cause was not the algorithm but the data feeding it: the colour/type features for its training pairs came from a classifier fitted on the same vehicles, so they were unrealistically accurate and the forest learned to over-trust them. Out-of-fold features fixed it. Finding this taught me to ask, for every feature, "would this value look the same on data the model has never seen?" — the same question behind identity-disjoint splits.

Combining weak evidence is powerful, but a learned combiner needs care. No single cue was reliable: plates disappear, colour labels are wrong a quarter of the time, appearance changes with viewpoint. Together they were robust. I also learned that a learned fusion is not automatically better than a transparent one: the hand-set weighted fusion beat the Random Forest with blurred plates, and monotonic constraints were needed so that more similarity could never mean a lower score.

Keeping a human in the loop is a design decision, not a weakness. Making the system return ranked candidates with the evidence for each, and requiring a human confirmation between the two stages, made the results explainable and made errors cheap to catch.

Reporting negative results honestly. The SOM pre-filter lowered accuracy. Reporting that clearly, together with the caveat that one modelling choice was informed by test queries, made the work more credible, not less.

3. Applying these skills in other situations and workplaces

- **Any multi-source matching problem** follows the same pattern as V-Trace: record linkage across databases, fraud detection across transactions, or matching products, documents or customers across systems. Each has several weak, partly missing signals that are stronger together, and benefits from a transparent baseline next to a learned model.
- **Leakage checks in any ML pipeline.** Stacked models, target encoding and features computed from labels are common in industry; out-of-fold features and grouped cross-validation are directly reusable.
- **Video analytics and smart-city systems.** The detection → tracking → description → retrieval chain transfers to traffic monitoring, parking and logistics yards, or retail footfall analysis, with privacy-preserving defaults such as ranked candidates instead of identity claims and access control.
- **Human-in-the-loop decision support.** Showing ranked options with their evidence, rather than a single automatic answer, is how I would design AI tools for any high-stakes setting, from investigations to quality inspection.
- **Engineering practice.** Small pull requests, automated tests and CI, cached processing and reproducible experiment scripts made a ten-day project manageable; I will use the same workflow on future team projects.